Presentation notes

A practical demonstration and viva guide

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Public project: https://github.com/ZekRock18/brain-tumor-segmentation

Live notebook: https://www.kaggle.com/code/zekrock/brain-tumor-seg-brats-africa

30-second introduction

“Our project evaluates two pretrained 3D brain-tumor segmentation models, UNet3D and UNETR, on real BraTS-Africa MRI scans. We ran both models on four patients using a Kaggle Tesla T4 GPU. The notebook displays MRI scans, reference masks and model predictions, and measures Dice overlap and HD95 boundary distance. We did not train or fine-tune the models.”

90-second explanation

“Each patient has four MRI modalities that provide complementary information. We normalize them using the original project's preprocessing and load its exact pretrained checkpoints. The models process overlapping 3D patches, which are blended into a full-volume prediction. We compare enhancing tumor, tumor core and whole tumor with the supplied labels. Our mean Dice values are 0.8546 for UNet3D and 0.8562 for UNETR. Mean HD95 values are 6.36 and 5.64 millimetres. Both models work on these examples, but four cases are too few to claim one is generally better. Our contribution is making the pretrained workflow run end to end on Kaggle, adapting the data loading and producing verified results and visualizations.”

Remember	Meaning
4 patients / 8 evaluations	Every selected patient was evaluated by both models
ET / TC / WT	Enhancing tumor / tumor core / whole tumor
Dice up; HD95 down	More overlap and smaller boundary error are better
No training	All reported predictions use released pretrained weights

Credit the work accurately

Say “we integrated and evaluated the pretrained models,” not “we trained these architectures.” Attribute the model code and checkpoints to the original project. Say “four-case demonstration,” not “validated clinical system” or “full benchmark reproduction.”

Live demonstration sequence

1. Show the completed run

Open results/executed.ipynb or the saved Kaggle version. Point to the GPU device output, strict weight loading and the final completion marker. Show the saved run rather than starting a fresh GPU execution during the presentation.

2. Explain the input

Open results/report.html. Show a four-modality preview: T1, contrast-enhanced T1, T2 and FLAIR. Explain that the target is a 3D tumor mask, not a yes/no cancer diagnosis.

3. Compare the segmentations

Show case 00047-000 first: the same MRI slice is shown with the supplied reference and each model's prediction. Explain the red/green/blue overlay legend. Open its GIF to move through the volume. The animation shows changing axial slices, not a rotating 3D mesh.

4. Discuss a difficult example

Show case 00170-000. UNet3D mean Dice is 0.8611 and UNETR is 0.9381, while their mean HD95 values are 14.18 and 6.75 mm. Use this to explain why visual review and boundary metrics accompany Dice. Other cases favor UNet3D; avoid choosing only flattering examples.

5. Close with the actual table

Show the four-patient summary and full CSV. State that the very small mean-Dice difference does not establish a general winner. The selected-case manifest, model hashes and executed notebook make the demonstration traceable.

If the internet or GPU is unavailable

Open START_HERE.html and results/report.html locally. The PNGs, GIFs, PDFs and CSVs work from the final folder without a new model run. The executed notebook preserves the earlier outputs. Do not claim that an offline report is a new live inference execution.

Viva questions and honest answers

Why use pretrained models?
They already contain learned parameters. The project demonstrates loading, inference and evaluation without the compute and data required to reproduce training.

How are the two models different?
UNet3D learns volumetric features through convolutions. UNETR uses a transformer encoder with a convolutional decoder. Both consume four MRI channels and output three tumor-region predictions.

What proves the checkpoints were used?
Strict state-dictionary loading, exact SHA-256 identities and the saved execution outputs. Speed or parameter count alone would not prove this.

Why sliding windows?
They allow full-size volumes to be processed using manageable 128-cubed patches. Overlap and Gaussian blending combine local predictions across the image.

What is ground truth?
The dataset's supplied reference segmentation. We compare predictions with that annotation; it is not generated by our model.

Are these unseen test patients?
Training-set overlap has not been independently checked. They are evaluation examples for this demonstration, not a guaranteed unseen test set.

Which model wins?
UNETR has a slightly higher mean Dice here; each model is better on two cases. Four patients do not support a broad superiority claim.

What remains to improve?
Confirm label semantics independently, audit overlap with training data, evaluate a larger predefined sample, report uncertainty and benchmark runtime under controlled conditions.

Did you implement training?
The upstream framework includes training code, retained for reference. Our demonstrated workflow is inference-only and produced no new trained model.

Before presenting: open the offline report, confirm the four GIFs play, keep the teacher PDF and results table ready, and retain the original final folder structure.